Response to Reviewers' Comments

We would like to thank all reviewers for reading our submission and offering constructive suggestions that have helped improve the manuscript. Their feedback has guided us to address critical gaps, and we will make comprehensive revisions, including supplementing independent module validations, expanding comparisons with state-of-the-art 3D Gaussian Splatting methods, adding detailed implementation,correcting minor errors and optimizing experimental presentation. Our detailed responses to each comment are provided below.

To Reviewer 1

1. *It is a significant drawback that the two proposed methods (EAD and GAA) were not tested independently in experiments 1 and 2. While these were addressed in experiment 3 (Ablation Study) for a single scene, the broader results lack this modular breakdown.*

Response: We appreciate your constructive comment on the independent validation of the EAD and GAA modules.To clarify the contribution and effectiveness of each component more clearly, we have reorganized and strengthened the ablation study with a more systematic modular breakdown. Specifically, we present comprehensive comparisons under a unified setup:Baseline only, Baseline + EAD,Baseline + GAA and Baseline + EAD + GAA.This structured ablation explicitly quantifies the individual performance improvements brought by EAD and GAA, as well as their synergistic effect when combined.The results clearly demonstrate that each module yields consistent and measurable gains across scenes, which sufficiently validates their respective effectiveness in a interpretable manner. The updated ablation study results are shown in Tables 1 & 2.

Table 1: Ablation experiments on the Tanks and Temples

Method-Metrics	PSNR↑	SSIM↑	LPIPS↓	#G↓
baseline	24.51	0.875	0.154	0.96
baseline + GAA	24.62	0.875	0.1581	0.84
baseline + EAD	25.12	0.888	0.135	1.54
baseline + EAD+ GAA	25.37	0.890	0.1381	0.96

Table 2: Ablation experiments on the Deep Blending

Method-Metrics	PSNR↑	SSIM↑	LPIPS↓	#G↓
baseline	29.67	0.909	0.220	2.54
baseline + GAA	29.86	0.913	0.217	1.39
baseline + EAD	29.95	0.914	0.212	1.92
baseline + EAD+ GAA	30.10	0.917	0.209	1.64

2. *Experiment 3 (Ablation Study) - Drawing conclusions or showcasing superiority based on a single image can be misleading.*

Response: Thank you for the constructive comment. To address the concern that ablation results on a single scene may be insufficient and potentially misleading, we have expanded the ablation study to multiple datasets instead of only the truck scene. We now perform comprehensive ablation experiments on both the Tanks and Temples dataset and the Deep Blending dataset. The cross-dataset validation ensures the effectiveness and generalization of EAD and GAA modules are reliably demonstrated, making

the conclusions more solid and credible. We have updated the corresponding results and descriptions in the revised manuscript.These experimental results are presented in Table 1 and Table 2.

3. *The paper lacks essential implementation details. Specifically, there is no clear explanation of how the threshold for densification was selected or if it remained constant across all diverse datasets. Furthermore, some mathematical formulations and parameters are not fully explained, leaving gaps in the method's reproducibility.*

Response:Response: We thank the reviewer for this valuable comment on implementation details and reproducibility. All hyperparameters in our method follow the default settings of the original 3DGS, including the densification threshold $\tau = 0.0002$, which is kept consistent across all datasets without modification. In contrast to the original scheme, our EAD module first collects Gaussian candidates whose gradients exceed the threshold, computes their importance weights using information entropy, and then determines densification probabilistically based on these weights. For the GAA module, we have reconstructed Equation (5) more clearly to better illustrate its working mechanism. Furthermore, we have added more detailed descriptions in key Sections 2.2 and 2.3 of the revised manuscript.

To Reviewer 2

1. *The main limitation of the paper is the lack of comparison with recent works that address similar objectives. Their method is only compared with the original 3DGS work from INRIA. The other methods shown in Table 1 do not even use Gaussian splats at all. In fact, Table 1 include works that were never mentioned in the text and are shown without proper reference (Plenoxels and INGP) This is not sufficient given that 3DGS is such a hot topic these days and many works were recently developed with similar objectives including pruning (e.g., LighGaussian, PUP-3DGS, CompactGS) and alternate densification strategies (e.g. AbsGS). Without such comparisons it is difficult to say if the proposed approach is relevant in such a competitive field.*

Response: We sincerely thank you for this constructive feedback. We fully agree that the initial comparison was not optimally aligned with the current landscape of 3DGS optimization, and we appreciate your insight that the comparison should focus on methods sharing the same core objective: reducing the number of Gaussian primitives rather than those focused on bit-level compression.

In response to this comment, we have completely re-conducted our experiments with a rigorously updated set of state-of-the-art baselines that are most relevant to our work. We have replaced the previous methods with several representative compaction methods, including TamingGS, Mini-splatting,etc. all of which are typical approaches focused on reducing Gaussian primitives while preserving visual fidelity.

We clarify the scope of our comparisons by distinguishing the underlying mechanisms of different 3DGS acceleration techniques.Our method and the newly selected baselines are both categorized as 3DGS compaction approaches, characterized by the shared goal of minimizing Gaussian primitives while maintaining rendering fidelity..By contrast, methods such as LightGaussian and EAGLES fall into the category of 3DGS compression.These approaches mainly reduce per-Gaussian storage overhead via techniques such as vector quantization and hashing, rather than decreasing the total number of Gaussians.In fact, they typically keep the full set of original Gaussians, or even increase the number of primitives before quantization. Our updated experiments in Table 3 now demonstrate that our method achieves superior performance specifically in the "compaction" regime. By focusing on this specific category, we provide a fairer and more rigorous comparison against the most relevant competitors in the field.

Table 3: **We present a revised comparative experiment in Table 3. Results for models marked with * are obtained by re-implementing their code on our local machines. The red and orange colorings denote the first and second ranked scores, respectively.The unit of "#G"is million.**

Datasets	Mip-NeRF360					Deep Blending					Tanks&Temples				
Method-Metrics	PSNR↑	SSIM↑	LPIPS↓	Mem↓	#G↓	PSNR↑	SSIM↑	LPIPS↓	Mem↓	#G↓	PSNR↑	SSIM↑	LPIPS↓	Mem↓	#G↓
3DGS*	27.16	0.802	0.220	638MB	2.70	29.99	0.916	0.215	504MB	2.13	24.20	0.865	0.165	343MB	1.45
AbsGS*	27.26	0.803	0.210	816MB	3.45	29.67	0.909	0.220	401.7MB	1.70	24.51	0.875	0.154	227.8MB	0.96
Mini-splatting*	27.65	0.815	0.219	752.7MB	3.18	29.80	0.910	0.221	592.9MB	2.60	24.65	0.872	0.168	351.1MB	1.49
TamingGS	27.79	0.822	0.205	-	3.31	30.14	0.907	0.235	-	2.81	24.04	0.851	0.170	-	1.84
Perceptual-GS	28.01	0.839	0.172	-	2.69	29.94	0.907	0.231	-	2.86	23.90	0.857	0.151	-	1.72
BogausS	28.82	0.844	0.171	-	3.21	31.25	0.915	0.229	-	2.80	25.99	0.872	0.170	-	1.83
DashGS	27.92	0.826	0.208	-	2.04	30.02	0.907	0.248	-	1.94	23.97	0.851	0.180	-	1.19
Ours (EAD-GS)	28.190	0.831	0.180	583MB	2.47	30.10	0.917	0.209	346MB	1.46	25.37	0.890	0.138	227MB	0.96

2. *The showed quantitative comparison reflects a RD trade-off rather than clear improvement. The authors say that they achieve lower memory usage and faster training but with a lower quality, suggesting a different operating point in the RD curve. It would be interesting to see reconstruction quality vs number of Gaussians when varying densification parameters.*

Response: Thank you for your insightful comment regarding the RD trade-off in our quantitative comparison.We acknowledge that 3DGS models do exhibit a certain degree of RD trade-off in some cases.Drastically reducing the number of Gaussian primitives inevitably degrades reconstruction quality, and in methods like DistillGS and LightGaussian, their performance metrics can even be inferior to the original 3DGS in certain scenes.

However, we would like to clarify that our proposed method is not merely a simple RD trade-off. In our updated experiments, we have adjusted the parameter design of the GAA module: we reduced the threshold for metric gain to 0.1, which limits the upper bound of the number of Gaussian primitives in each scene. This adjustment makes the densification process less aggressive compared to our initial experimental setup. The updated experimental results demonstrate that our method can maintain the rendering quality of the baseline model while using fewer Gaussian primitives—though it requires a slightly larger

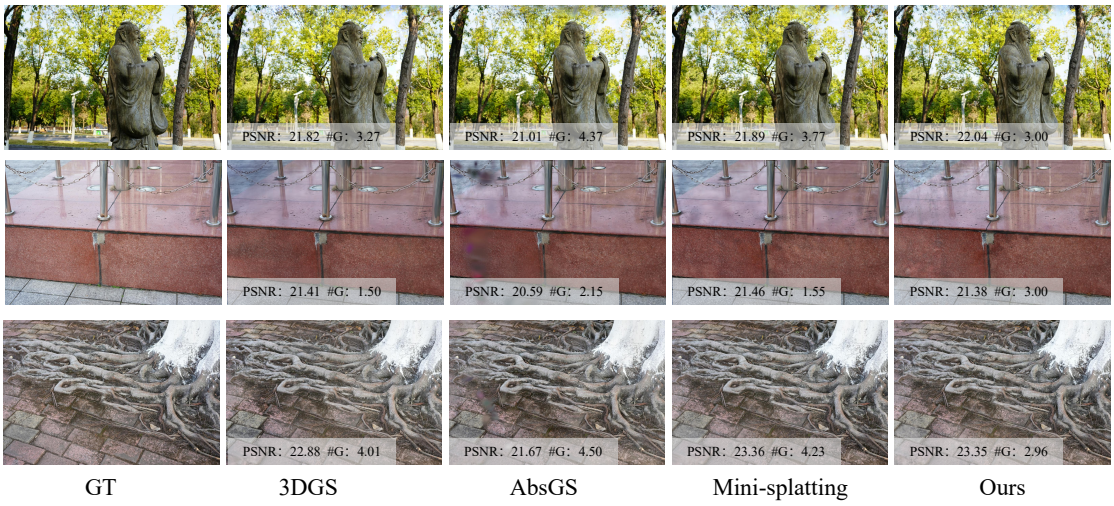

Figure 1: **Comparison of test results between our method and others on self-constructed datasets**

number of Gaussians than our previous parameter setting, this adjustment effectively avoids the trade-off between quality and Gaussian count, achieving a more optimal balance that is not a pure RD trade-off.

3. *It is said that for entropy calculation, image is converted to grayscale, it would be good to precisely specify which conversion method was used.*

Response: Thank you for your valuable suggestion and careful review of our manuscript.The grayscale image is obtained using the standard RGB-to-grayscale conversion formula :

$$Gray = 0.299R + 0.587G + 0.114B$$

These coefficients correspond to the luminance perception weights of the human visual system, where green contributes the most and blue the least to perceived brightness.

4. *On 3.2 the authors claim the proposed method provides compression of the trained method. As commented in 3DGS.zip, compression is not the proper term here as the reduction of size comes from the reduction of the number of Gaussians, it is rather a compaction method (and should be compared with such methods).*

Response: We appreciate your careful remark on the terminology. We agree that "compression" is not an accurate term, as the model size reduction in our work stems directly from decreasing the number of Gaussians. In the revised manuscript, we have replaced "compression" with "compaction" throughout the text, especially in Section 3.2, and frame our method as a Gaussian compaction approach with fair comparisons to other compaction-oriented 3DGS methods.

5. *The paper talks a lot about reducing the number of primitives but the information on the actual number on their approach vs the original 3DGS is never mentioned.*

Response: We sincerely apologize for not clarifying the relationship between the number of Gaussian primitives and model memory size in the original manuscript. In our comparative experiments, we reported the model memory footprint as an indicator, which is actually equivalent to the number of Gaussian primitives. As the number of Gaussian primitives decreases, the model memory footprint decreases proportionally. Interestingly, the ratio of memory size to primitive count remains consistent at approximately 2.365 across all experiments, which further confirms the equivalence between these two indicators.

To avoid such misunderstandings in the revised manuscript, we have supplemented both indicators in all experimental tables: we now report both the model memory footprint and the number of Gaussian primitives, ensuring that the experimental results are more transparent and easier to understand for

readers.

6. *Minor corrections: - Caption of Fig. 1 "overhead.and simultaneously..." - First sentence of the second paragraph of introduction should be inverted: "The component of 3DGS which automatically regulates... is ADC"- The term "vanilla 3DGS" is not ideal, it sounds informal and slightly dismissive for such an important work.*

Response: Thank you for these comments. We have corrected these mistakes accordingly in the revised manuscript. Also, we have made efforts to modify the overall structure of the manuscript and improve the presentation style.

To Reviewer 3

1. *The main reason why I cannot recommend acceptance for this paper is the fact that the experimental validation is lacking in many aspects. The authors do not compare with other densification modules; the comparison is mainly done with results published in the 2023 paper that introduced GS. Moreover, the results are misleading: the authors say that "this metric [memory] surpasses 50% on the Mip- NeRF360 and Deep Blending datasets", but that's only with respect to vanilla 3DGS. According to the table, their method uses 285MB while INGP-Base, INGP-Big, and Mip-NeRF360 use significantly less memory (Mip-NeRF360 uses 8.6MB, a fraction of the proposed method), with comparable quality. Maybe time could be an advantage, but the authors do not show the time gain with respect to NeRF methods. If gains were shown with respect to other 3DGS methods, at least that would be a validation; but as I mentioned before, no comparison is made with state of the art gaussian densification methods.*

Response: We sincerely thank you for your insightful and constructive feedback. We fully agree with your observations regarding the limitations of our initial experimental design: our comparisons were inadequate, as we focused primarily on the original 3DGS without incorporating state-of-the-art Gaussian densification modules, and our description of memory performance was misleading in failing to clarify our method's time consumption advantages relative to NeRF-based approaches. Such comparisons of different evaluation metrics across distinct architectural methods are unreasonable and confusing.We sincerely apologize for any confusion this insufficient explanation may have caused.

To address these concerns, we have completely redesigned our experimental setup, removing all cross-architecture comparisons and focusing exclusively on recent point-based 3DGS methods for our comparisons.We intentionally select baselines aligned with our method's core principle: reducing memory footprint through control of Gaussian primitive count. This contrasts with approaches such as LightGaussian and CompactGS, which instead modify the bitwidth or storage format of individual Gaussians. As such, we have selected representative state-of-the-art Gaussian densification strategies, including SteepGS, GaussianSPA, and DashGS, all of which align closely with our core objective of optimizing Gaussian count for memory efficiency. This intentional selection ensures our comparisons are fair, relevant, and targeted, allowing us to rigorously demonstrate the effectiveness of our proposed densification control approach against peer methods with the same fundamental goal.The revised comparative experiments are presented in Table 3 and Figure 1.

Our updated experimental results validate the superiority of our method in Gaussian densification control and memory efficiency among these peer 3DGS approaches.

2. *Equations should be better explained. It is not clear to me what threshold is used for Gaussian pruning since the authors never state it. Equation (5) is equally baffling: why is the total entropy of the training dataset used? and how does it affect performance when seeing new datasets?*

Response: We sincerely appreciate your valuable feedback on the clarity of the formula. In the latest revised version, we have provided a more detailed explanation for the formula. Specifically, the original form of formula (5) in the initial draft is:

$$Num = \lambda \cdot H(p) \cdot SFM_{init} \tag{5}$$

Our initial intention was to use the global entropy $H(p)$ of the training dataset to quantify the richness of the scene, and then set differentiated densification rates for scenes of different complexities. However, for a specific dataset, global entropy is a constant value. Therefore, in the revised manuscript, we have rephrased formula (5) where the global entropy $H(p)$ constant is retained in the formula as a regular coefficient. Currently, we focus on elaborating on the key growth factors λ, which are defined as follows:

$$\lambda = K(\sqrt{\frac{x-500}{14000} - \frac{x-\triangle x-500}{14000}})$$

where x denotes the number of training iterations,λ is a variable closely related to the number of densification iterations. It follows a bounded and gradually converging function law - in the early stage of densification, we quickly supplement Gaussian elements; In the later stage, the rate of Gaussian growth is slowed down. This design ensures more precise and efficient control of Gaussian quantities. In addition, in the revised manuscript, we also clearly defined the threshold for Gaussian pruning.

3. *-The presentation is quite poor, an English proof-reading is needed-The references are added to strange sentences. I.e., "Papantonakis et al. [13] proposed stripping redundant attributes from each primitive to reduce the burden of parameter optimization [14–17]" how is [14-17] relevant when talking about [13]?*

Response: Thank you for these comments. We have corrected these mistakes accordingly in the revised manuscript. Also, we have made efforts to modify the overall structure of the manuscript and improve the presentation style.